TEXORA · LECTURE SHEET

Basic Principles of Electrical Machines

The D.C. generator and motor — construction, working principle, EMF equation and speed control; the induction motor and its slip–torque behaviour; and the transformer — construction, EMF equation and the condition for maximum efficiency.

EEE-2203 · BASICS OF ELECTRICAL & ELECTRONICS ENGINEERING · TOPIC 2 · LEVEL 2 TERM 2

1. The common principle

Every rotating electrical machine is an energy converter working on two laws of electromagnetism. **Faraday's law:** when the flux linking a conductor changes, an EMF is induced in it, $e = -N\,d\Phi/dt$; mechanically, a conductor of length l moving at speed v across a field B has $e = Blv$ induced in it. **The force law:** a conductor carrying current I in a field B experiences a force $F = BIl$. A **generator** uses the first effect (mechanical power in, electrical power out); a **motor** uses the second (electrical in, mechanical out). The same machine can usually do both.

2. The D.C. generator: construction

PART	FUNCTION
Yoke (frame)	Cast-iron or steel body: mechanical support and the return path for the magnetic flux.
Field poles and field winding	Salient poles bolted to the yoke, carrying coils fed with D.C. (the field/exciter current); they produce the main flux Φ per pole.
Armature core	Laminated steel cylinder on the shaft, slotted to hold the armature conductors; laminations reduce eddy-current loss.
Armature winding	Copper conductors in the slots, connected in **lap** winding (parallel paths A = P) or **wave** winding (A = 2), where P is the number of poles.
Commutator	A ring of copper segments, insulated from each other, rotating with the armature; it reverses each coil's connection at the right instant so the external EMF stays unidirectional.

Brushes	Carbon blocks pressing on the commutator; they collect the current and deliver it to the external circuit.

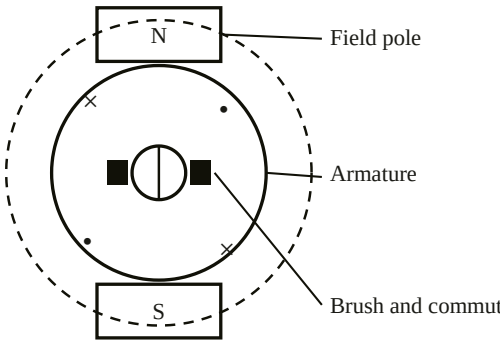


Fig. 1 — Simplified cross-section of a two-pole D.C. machine. × and • show conductor current directions on the two sides of the armature.

3. Working principle and the EMF equation

The prime mover rotates the armature. Each conductor cuts the pole flux, so an alternating EMF is induced in the armature coils; the commutator and brushes rectify it mechanically, giving a steady D.C. voltage at the terminals. If Φ is the flux per pole (Wb), P the number of poles, Z the total armature conductors, N the speed in rpm and A the number of parallel paths through the winding, the average EMF generated is

$$E_g = (P\Phi N/60) \times (Z/A) \quad \text{volts}$$

WORKED EXAMPLE — GENERATOR EMF

A 4-pole D.C. generator has Z = 480 armature conductors, flux per pole Φ = 0.02 Wb, and runs at N = 1000 rpm with a wave winding (A = 2).

$E_g = (4 \times 0.02 \times 1000/60) \times (480/2) = (80/60) \times 240 =$ **320 V**.

With a lap winding instead (A = P = 4), $E_g = (80/60) \times 120 = 160$ V — lap winding gives more parallel paths, so less voltage but more current capacity.

4. Motors: construction and working principle

A **D.C. motor** has the same construction as the D.C. generator. Supplied with D.C., the armature conductors carry current in the field flux and experience forces F = BIl that produce torque in one rotational direction; the commutator keeps reversing the coil currents so the torque never changes sign.

As the armature speeds up it also acts as a generator, producing a **back EMF** $E_b = (P\Phi N/60)(Z/A)$ opposing the supply: $V = E_b + I_aR_a$. The back EMF is the motor's self-regulator — it limits the armature current to $I_a = (V - E_b)/R_a$, which is why a D.C. motor draws a dangerously large current at the instant of starting ($E_b = 0$) and needs a starter.

The **three-phase induction motor**, the workhorse of the textile mill, has a stator wound for three phases and a rotor that is either a squirrel cage (aluminium bars shorted by end rings) or wound with slip rings. The three-phase stator currents create a field rotating at the **synchronous speed** $N_s = 120f/P$ rpm. The rotating field cuts the rotor conductors, induces rotor currents, and the reaction produces torque — the rotor follows the field but always a little behind it.

5. Speed control of motors

For a D.C. motor, $N \propto E_b/\Phi$, giving three classical methods:

- **Armature voltage control:** varying the armature supply voltage varies speed smoothly (below base speed) — the basis of modern electronic drives.
- **Field (flux) control:** weakening the field with a rheostat in the field circuit raises speed (above base speed), at reduced torque.
- **Armature resistance control:** a series rheostat drops voltage across itself; simple but wasteful, used only for short-time duty.

For an induction motor, speed follows $N_s = 120f/P$ and the slip: control methods are **frequency control** (with voltage varied in proportion, V/f control — the modern standard), **pole changing** (stepped speeds), **stator voltage control** (limited range, e.g. fans), and, for wound-rotor machines, **rotor resistance control**.

6. Slip and the torque of an induction motor

The fractional lag of the rotor behind the rotating field is the **slip**:

$$s = (N_s - N)/N_s, \quad \text{rotor EMF frequency } f_r = s\,f$$

At standstill $s = 1$ and the rotor is like a transformer secondary at full frequency; at synchronism $s = 0$, no EMF is induced and no torque is produced — an induction motor can never quite reach N_s. The torque developed is

$$T = k \cdot s\,E_2^2\,R_2 / (R_2^2 + (sX_2)^2)$$

where E_2 is the rotor standstill EMF, R_2 the rotor resistance and X_2 the rotor standstill reactance (k a machine constant). At small slip the denominator is nearly R_2^2, so $T \approx (k\ E_2^2/R_2)\cdot s$ — torque is proportional to slip, the stable working region. Maximum torque occurs at slip $s_{max} = R_2/X_2$.

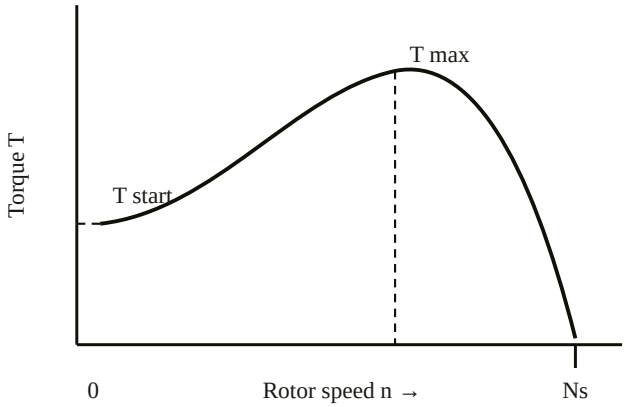


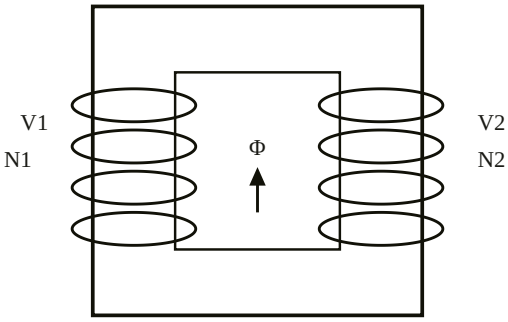
Fig. 2 — Torque–speed characteristic of an induction motor: torque is zero at synchronous speed Ns and maximum at the pull-out slip.

WORKED EXAMPLE — SLIP

A 4-pole induction motor on a 50 Hz supply has $N_s = 120 \times 50/4 = 1500$ rpm. If the rotor runs at 1440 rpm: $s = (1500 - 1440)/1500 =$ **0.04, i.e. 4%**, and the rotor currents alternate at $f_r = sf = 0.04 \times 50 =$ **2 Hz**.

7. The transformer: construction and principle

A transformer transfers A.C. power between circuits at different voltage levels through a common magnetic flux — there is no electrical connection and no moving part. A laminated steel **core** (core-type or shell-type) carries two windings: the **primary** (N_1 turns) connected to the supply and the **secondary** (N_2 turns) connected to the load; in large units the whole assembly sits in an oil-filled tank that insulates and cools it. Alternating primary current sets up an alternating flux Φ in the core; by Faraday's law this flux induces an EMF in both windings (mutual induction). If the tank oil heats and expands, a conservator and a Buchholz relay provide protection (Topic 4).



Primary (left) and secondary (right) on a common core

Fig. 3 — Two-winding transformer: the alternating core flux Φ links both windings.

For a sinusoidal flux of maximum value Φ_m, the RMS EMF induced in a winding of N turns is

$$E = 4.44\, f\, N\, \Phi_m \quad \cdot \quad E_1/E_2 = N_1/N_2 = V_1/V_2 \text{ (ideal)}$$

WORKED EXAMPLE — TRANSFORMER EMF

A transformer primary has $N_1 = 400$ turns and is connected to 230 V, 50 Hz. $\Phi_m = E_1/(4.44\, f\, N_1) = 230/(4.44 \times 50 \times 400) = 230/88\,800 =$ **2.59×10^{-3} Wb ≈ 2.6 mWb**. With $N_2 = 40$ turns, $V_2 = 230 \times 40/400 =$ **23 V** — a 10 : 1 step-down.

8. Transformer losses and maximum efficiency

Losses are of two kinds: **iron (core) loss** — hysteresis plus eddy currents — which is practically constant at all loads because it depends on voltage and frequency; and **copper loss** I^2R in the windings, which grows with the square of the load current. Efficiency η = output/(output + losses). Efficiency is maximum at the load where the two losses are equal:

Maximum efficiency when variable (copper) loss = constant (iron) loss, i.e. $I^2R = P_{iron}$

Distribution transformers, which are energised all day but lightly loaded much of the time, are designed so this point falls at about half to two-thirds of full load.

9. Key terms

- **Armature:** the rotating part of a D.C. machine carrying the winding in which EMF is induced (or which produces torque).
- **Commutator:** the segmented ring that mechanically rectifies the armature EMF (generator) or reverses coil currents (motor).

- **Back EMF:** the EMF a motor's armature generates opposing the supply; it regulates the armature current.
- **Synchronous speed:** $N_S = 120f/P$, the speed of the rotating field of an A.C. machine.
- **Slip:** the fractional difference between synchronous and rotor speed, $s = (N_S - N)/N_S$.
- **Mutual induction:** EMF induced in one winding by the changing flux produced by another winding on the same core.
- **Iron loss / copper loss:** the roughly constant core loss (hysteresis + eddy current) and the load-dependent I^2R winding loss.

10. Key points

- Generator action is Faraday's law; motor action is the force on a current-carrying conductor. The D.C. machine does both, separated only by the direction of energy flow.
- $E_g = (P\Phi N/60)(Z/A)$: lap winding ($A = P$) suits high current, wave winding ($A = 2$) suits high voltage.
- A D.C. motor's back EMF limits its running current but is zero at starting — hence starters.
- Induction-motor torque is proportional to slip at working slips, maximum at $s = R_2/X_2$, and zero at synchronism.
- Transformer EMF equation $E = 4.44\ f\ N\ \Phi_m$ fixes the design: volts per turn are set by f and Φ_m.
- Transformer efficiency peaks where copper loss equals iron loss.

11. Summary

Electrical machines convert between mechanical and electrical energy through the magnetic field. The D.C. machine uses a commutator to obtain unidirectional voltage or torque, with its EMF fixed by Φ, Z, N, P and the winding paths. The induction motor replaces the commutator with a rotating field and a slip-dependent torque, controlled most flexibly by varying the supply frequency. The transformer, static and highly efficient, couples two windings by a common core flux, stepping voltage in proportion to turns, and reaches maximum efficiency when its variable copper loss equals its constant iron loss.

12. Practice

Answer in your own words or by calculation; brief answers follow. No multiple-choice questions.

1. A 6-pole induction motor runs on a 50 Hz supply with the rotor at 960 rpm. Find the synchronous speed and the slip.

2. A transformer winding of 500 turns is designed for 50 Hz with a maximum core flux of 0.004 Wb. What voltage is it rated for?
3. Using the data of the worked example in Section 3, state the EMF if the same generator is lap-wound, and explain in one line why it changes.
4. Why must a starter be used with a D.C. motor, and what quantity does the starter limit?
5. State the condition for a transformer's maximum efficiency, and explain why a distribution transformer is designed to reach it below full load.

Answer key

1. $N_s = 120 \times 50/6 = 1000$ rpm. $s = (1000 - 960)/1000 = 0.04 = 4\%$.
2. $E = 4.44\, f\, N\, \Phi_m = 4.44 \times 50 \times 500 \times 0.004 = 444$ V.
3. 160 V. Lap winding has $A = P = 4$ parallel paths instead of 2, so the conductors in series per path halve and the EMF halves (current capacity doubles).
4. At standstill the back EMF is zero, so the armature current $(V - 0)/R_a$ would be many times the rated value because R_a is very small. The starter inserts resistance in series to limit this starting current, cutting it out as speed and back EMF build up.
5. Maximum efficiency occurs when copper loss equals iron loss ($I^2R = P_{iron}$). A distribution transformer is energised for 24 hours (iron loss runs continuously) but carries heavy load for only part of the day, so its all-day efficiency is best when the peak occurs at the typical load, about 50–70% of full load.
